# THE COLLISION PRIME NUMBER THEOREM

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Abstract. For a prime $p$ not dividing a positional base $b$, the collision count $C(b \bmod p)$ records the number of residues $r$ for which the digit function $\delta(r) = \lfloor br/p \rfloor$ satisfies $\delta(r) = \delta(br \bmod p)$. The signed collision count $S(p) = C(b \bmod p) - \lfloor (p-1)/b \rfloor$ is an integer determined by $p \bmod b^2$, taking values in $[-(b-1),\ b-2]$ with the antisymmetry $S(a) + S(b^2-a) = -1$. The $\varphi(b^2)$ values form the *collision periodic table*, a finite invariant of the base.

The centered collision deviation $S^\circ(p)$ is a linear combination of nontrivial odd Dirichlet characters modulo $b^2$ (primitive odd for prime bases). From this decomposition we prove the *collision prime number theorem* ($\sum S^\circ(p) = O(x \exp(-c\sqrt{\log x}))$, effective for prime bases), the *collision variance* ($\sum |S^\circ|^2 \sim \sigma^2 x/\log x$), and $L$-value extraction: for prime bases, the periodic table encodes $|L(1,\chi)|$ for all primitive odd characters modulo $b^2$ (verified computationally for composite bases through $b = 60$).

For base 10, we prove the diagonal–Bernoulli identity $|S_G(\chi)| = |B_{1,\overline{\chi}}|$ (with base 6 by the same method and base 12 verified computationally), giving $|\widehat{c}(\chi)|^2 \propto |L(1,\chi)|^4$: the collision weights are a fourth power of the $L$-value. The weights live in $\mathbb{Q}(\sqrt{5})$ for base 10, where $\sqrt{5}$ arises from the 5th roots of unity in the character theory of $(\mathbb{Z}/25\mathbb{Z})^*$.

## 1. Introduction

The digit function $\delta(r) = \lfloor br/p \rfloor$ assigns to each residue $r \in \{1, \ldots, p-1\}$ a digit $d \in \{0, \ldots, b-1\}$. Multiplication by $g \bmod p$ permutes the residues. The collision count $C(g) = \#\{r : \delta(r) = \delta(gr \bmod p)\}$ measures how many residues share a digit with their image under this permutation. It encodes the interaction between the additive structure of the bins and the multiplicative action of $g$.

The preceding papers [1, 2, 3] established that the collision count decomposes into Dirichlet characters modulo $b^2$: the collision transform expands $S^\circ(p)$ over odd characters (primitive odd for prime bases),

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and the decomposition theorem [3] factors each coefficient as $\widehat{c}(\chi) = -B_{1,\overline{\chi}} \cdot \overline{S_G(\chi)}/\phi(m)$. (The decomposition is proved for prime bases in [3]; the Fourier analysis extends to all bases, and the identity is verified computationally for composite bases through $b = 60$.)

This paper proves the consequences of that decomposition. The collision PNT (Section 3) is the starting point. The destination is the fourth-moment identity $|\widehat{c}(\chi)|^2 \propto |L(1,\chi)|^4$ (Corollary 5.5), proved through a new algebraic identity between the diagonal character sum and the first Bernoulli number (Theorem 5.3).

## 2. The Periodic Table

The collision count at the specific multiplier $g = b$ has a closed form. Write $p = b^2 Q_2 + a$ with $0 < a < b^2$ and $\gcd(a, b) = 1$. Define

$$f(a) = \#\{d \in \{0, \ldots, b-1\} : (b+1)da \bmod b^2 \geq b^2 - a\}. \tag{1}$$

**Proposition 2.1** (Collision formula)**.** *For a prime $p > b^2$ with $p \nmid b$:*

$$C(b \bmod p) = b \cdot Q_2 + f(a) - 1,$$

*where $Q_2 = \lfloor p/b^2 \rfloor$ and $a = p \bmod b^2$. The slice decomposition of* [1] *gives $b$ intervals of length $p/b^2$, each contributing $Q_2$ or $Q_2+1$ collisions. The count $f(a)$ records which intervals gain the extra collision; the $-1$ excludes $r = 0$.*

The signed collision count $S(p) = C(b \bmod p) - \lfloor (p-1)/b \rfloor$ satisfies $S = f(a) - 1 - \lfloor (a-1)/b \rfloor$, depending only on $a = p \bmod b^2$.

**Theorem 2.2** (Finite determination)**.** *For any base $b \geq 2$ and prime $p > b^2$ with $p \nmid b$, the signed collision count $S(p)$ is an integer determined by $p \bmod b^2$.*

*Proof.* $S(p) = f(a) - 1 - \lfloor (a-1)/b \rfloor$ by Proposition 2.1, where $a = p \bmod b^2$. □

This reduces the study of collisions across all primes to a combinatorial problem on a finite group.

**Definition 2.3.** The *collision periodic table* is the function $\mathcal{T} \colon (\mathbb{Z}/b^2\mathbb{Z})^* \to \mathbb{Z}$ defined by $\mathcal{T}(a) = S(p)$ for any prime $p \equiv a \pmod{b^2}$ with $p > b^2$.

**Theorem 2.4** (Antisymmetry)**.** *For every base $b \geq 3$ and every coprime $a \bmod b^2$:*

$$\mathcal{T}(a) + \mathcal{T}(b^2 - a) = -1.$$

*Proof.* Three identities.

*Complementary counting.* Let $r_d = (b{+}1)da \bmod b^2$ for $d = 0, \ldots, b{-}1$. For $d \geq 1$: $r_d \neq 0$ (since $\gcd(a(b{+}1), b^2) = 1$). The complement $b^2 - r_d = (b{+}1)d(b^2{-}a) \bmod b^2$. The condition $r_d \geq b^2 - a$ for $f(a)$ and the condition $b^2 - r_d \geq a$ for $f(b^2{-}a)$ are complementary: every $d \geq 1$ satisfies exactly one, except when $r_d = b^2 - a$ (both). Including $d = 0$ ($r_0 = 0$, neither): $f(a) + f(b^2{-}a) = b$.

*Floor identity.* $(a{-}1) + (b^2{-}a{-}1) = b^2 - 2$, with remainders $R + R' = b - 2 < b$ modulo $b$, so $\lfloor (a{-}1)/b \rfloor + \lfloor (b^2{-}a{-}1)/b \rfloor = b - 1$.

*Assembly.* $\mathcal{T}(a) + \mathcal{T}(b^2{-}a) = [f(a) + f(b^2{-}a)] - 2 - [\lfloor (a{-}1)/b \rfloor + \lfloor (b^2{-}a{-}1)/b \rfloor] = b - 2 - (b{-}1) = -1$. □

**Corollary 2.5.** *$\mathcal{T}$ takes values in $[-(b{-}1),\ b{-}2]$. Exactly $\varphi(b^2)/2$ classes are negative. The mean of $\mathcal{T}$ is $-1/2$.*

*Proof.* $f(a) \geq 1$ because $d = b{-}1$ always contributes, so $\mathcal{T}(a) \geq -(b{-}1)$. By antisymmetry, $\mathcal{T}(a) \leq b - 2$. Each pair $(a, b^2{-}a)$ sums to $-1 < 0$, so every pair contributes exactly one negative member. □

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | . | 0 | . | +2 | . | . | . | 0 | . | +8 |
| 1 | . | −1 | . | −1 | . | . | . | +1 | . | −1 |
| 2 | . | 0 | . | −2 | . | . | . | +6 | . | 0 |
| 3 | . | −1 | . | −1 | . | . | . | −3 | . | −1 |
| 4 | . | −4 | . | 0 | . | . | . | −2 | . | 0 |
| 5 | . | −1 | . | +1 | . | . | . | −1 | . | +3 |
| 6 | . | 0 | . | +2 | . | . | . | 0 | . | 0 |
| 7 | . | −1 | . | −7 | . | . | . | +1 | . | −1 |
| 8 | . | 0 | . | −2 | . | . | . | 0 | . | 0 |
| 9 | . | −9 | . | −1 | . | . | . | −3 | . | −1 |

TABLE 1. The collision periodic table in base 10. Row is the tens digit, column is the units digit of $p \bmod 100$. Dots mark non-coprime classes. For any prime $p > 100$, the entry gives $S = C(10 \bmod p) - \lfloor (p{-}1)/10 \rfloor$.

## 3. The Collision Prime Number Theorem

The centered deviation $S^\circ = S - \overline{S}_R$ removes the class mean (which depends on the last digit $R = (p{-}1) \bmod b$). By Fourier inversion on

$(\mathbb{Z}/b^2\mathbb{Z})^*$:

$$S^\circ(p) = \sum_{\chi \neq \chi_0} \widehat{c}(\chi)\, \chi(p), \tag{2}$$

a sum over nontrivial characters modulo $b^2$. The antisymmetry $S^\circ(m-a) = -S^\circ(a)$ kills even characters; the centering kills characters whose conductor divides $b$ [2]. For prime bases, the surviving characters are precisely the primitive odd characters [3].

**Theorem 3.1** (Collision PNT)**.** *For any base $b \geq 2$:*

$$\sum_{\substack{p \leq x \\ p \nmid b}} S^\circ(p) = O\Big(x \exp\Big(-c\sqrt{\log x}\Big)\Big),$$

*where $c > 0$ depends on $b$ (effective for prime $b \geq 3$ by Theorem 3.2; effective for composite bases verified through $b = 60$).*

*Proof.* By (2), the sum equals $\sum_{\chi \neq \chi_0} \widehat{c}(\chi) \sum_{p \leq x} \chi(p)$, a finite sum (at most $\varphi(b^2) - 1$ terms) using only Fourier inversion on $(\mathbb{Z}/b^2\mathbb{Z})^*$, valid at every base. Each character sum satisfies $\sum_{p \leq x} \chi(p) = O(x \exp(-c_1\sqrt{\log x}))$ [5] for any nontrivial $\chi$ at fixed modulus $b^2$. For prime bases $b \geq 3$: the surviving characters are primitive odd (Theorem 3.2 excludes real primitive characters), so $c_1$ is effective. For composite bases: effectiveness holds provided no real character survives the centering, which is verified for all bases through $b = 60$. □

**Theorem 3.2** (Siegel immunity)**.** *For $b \geq 3$, no real primitive character modulo $b^2$ exists. For prime bases, where the surviving characters are all primitive, $c$ in Theorem 3.1 is effective.*

*Proof.* A quadratic character modulo $p^{2a}$ ($p$ odd) has conductor $p$, not $p^{2a}$: it is the lift of the Legendre symbol. At $p = 2$ with $2a \geq 4$, the conductor is at most 8. Since $b^2$ is a perfect square and $b \geq 3$, every quadratic character modulo $b^2$ is imprimitive. □

**Theorem 3.3** (Collision variance)**.** $\displaystyle\sum_{\substack{p \leq x \\ p \nmid b}} |S^\circ(p)|^2 = \sigma^2 \cdot \frac{x}{\log x} + O\left(\frac{x}{\log^2 x}\right),$

*where $\sigma^2 = \sum_\chi |\widehat{c}(\chi)|^2$.*

*Proof.* $S^\circ(p)$ depends only on $p \bmod b^2$, so the sum decomposes over residue classes. The PNT for arithmetic progressions with fixed modulus gives $\pi(x; b^2, a) = x/(\varphi(b^2)\log x) + O(x/\log^2 x)$. Summing and applying Parseval: $\sigma^2 = (1/\varphi(b^2)) \sum_a |S^\circ(a)|^2 = \sum_\chi |\widehat{c}(\chi)|^2$. □

The collision deviations cancel in the first moment (the PNT) but persist in the second moment (the variance). In base 10: $\sigma^2 \approx 7.26$, giving typical fluctuations $|S^\circ| \approx 2.7$.

## 4. $L$-Value Extraction

The periodic table encodes $L$-function magnitudes.

**Theorem 4.1** ($L$-value extraction)**.** *For prime bases $b$ (and computationally verified for composite bases through $b = 60$), the magnitudes $|L(1,\chi)|$ for all primitive odd characters modulo $b^2$ are determined by the collision periodic table via:*

1. *Fourier inversion of $\mathcal{T}(a)$ on $(\mathbb{Z}/b^2\mathbb{Z})^*$ recovers $\widehat{c}(\chi)$.*
2. *The diagonal sum $S_G(\chi) = \sum_{n\in G}[\overline{\chi}(n+1) - \overline{\chi}(n)]$ is computed from the diagonal set $G$ [3] (a finite computation depending on the base, not on primes).*
3. *The decomposition theorem gives $|B_{1,\overline{\chi}}| = \phi(m)\,|\widehat{c}(\chi)|/|S_G(\chi)|$, and the classical formula [5] converts to the $L$-value: $|L(1,\chi)| = (\pi/b)\,|B_{1,\overline{\chi}}|$.*

*The system is overdetermined ($\phi(b^2)$ table entries, $\phi(b^2)/2$ character coefficients), providing verification at each step.*

This is a new instance of an old pattern. Dirichlet's class number formula extracts $|L(1,\chi)|$ from the arithmetic of ideal classes through a formula involving $\pi$. The collision periodic table does the same from the arithmetic of digit collisions. In both cases, a finite object encodes analytic information that is, in principle, determined by the infinite distribution of primes.

| $k$ | $\lvert\widehat{c}\rvert$ | $\lvert B_{1,\overline{\chi}}\rvert$ | $\lvert L(1,\chi_k)\rvert$ | $\frac{\lvert\widehat{c}\rvert}{\lvert B_{1,\overline{\chi}}\rvert\cdot\lvert L\rvert}$ |
|---|---|---|---|---|
| 2 | 0.462 | 4.298 | 1.350 | 0.0796 |
| 4 | 0.362 | 3.804 | 1.195 | 0.0796 |
| 6 | 0.238 | 3.087 | 0.970 | 0.0796 |
| 8 | 0.138 | 2.351 | 0.739 | 0.0796 |
| 12 | 0.138 | 2.351 | 0.739 | 0.0796 |
| 14 | 0.238 | 3.087 | 0.970 | 0.0796 |
| 16 | 0.362 | 3.804 | 1.195 | 0.0796 |
| 18 | 0.462 | 4.298 | 1.350 | 0.0796 |

Table 2. $L$-value extraction in base 10. The ratio $|\widehat{c}|/(|B_{1,\overline{\chi}}| \cdot |L|) = 1/(4\pi)$ for all 8 primitive odd characters modulo 100.

## 5. The Diagonal–Bernoulli Identity

The single constant $1/(4\pi)$ in Table 2 has an algebraic explanation.

**Lemma 5.1** (Bernoulli–$L$ identity [5])**.** *For a primitive odd character $\chi$ modulo $m$: $|B_{1,\overline{\chi}}| = (b/\pi)\,|L(1,\chi)|$.*

**Corollary 5.2** ($L$-encoding)**.** $|\widehat{c}(\chi)| = (b/\pi\varphi(m))\,|L(1,\chi)| \cdot |S_G(\chi)|$.

The collision weight factors into a multiplicative part ($|L(1,\chi)|$, encoding the class number) and an additive part ($|S_G(\chi)|$, encoding the bin geometry of the digit function). The question is whether these two factors are related.

**Theorem 5.3** (Diagonal–Bernoulli identity)**.** *For all 8 primitive odd characters $\chi$ modulo 100 (base 10) and all 2 primitive odd characters modulo 36 (base 6):*

$$|S_G(\chi)| = |B_{1,\overline{\chi}}|.$$

**Observation 5.4.** The same identity holds for all 8 primitive odd characters modulo 144 (base 12), verified computationally to full precision.

*Proof.* Write $m = 100 = 4 \times 25$ and decompose each primitive odd character as $\chi = \chi_4 \times \psi$ where $\psi = \chi_{25}^k$ with $k \in \{2,4,6,8,12,14,16,18\}$.

**Step 1: CRT decomposition of $B_1$.** Using the Chinese Remainder Theorem with $a = 25a_1 + 76a_2 \bmod 100$, $a_1 \in \{1,3\}$, $a_2 \in (\mathbb{Z}/25\mathbb{Z})^\times$: the linear terms vanish by character orthogonality, and $B_{1,\overline{\chi}}$ reduces to:

$$B_{1,\overline{\chi}} = -\sum_{a_1\in\{1,3\}} \sum_{a_2} \left\lfloor \frac{25a_1 + 76a_2}{100} \right\rfloor \chi_4(a_1)\,\overline{\psi}(a_2).$$

**Step 2: The collision indicator.** Since $\chi_4(1) = 1$ and $\chi_4(3) = -1$, the sum over $a_1$ reduces to a floor difference: $\lfloor (75 + 76a_2)/100 \rfloor - \lfloor (25 + 76a_2)/100 \rfloor$. This equals 1 when $\{76a_2/100\} \in [1/4, 3/4)$ and 0 otherwise. The set $C = \{a_2 \in (\mathbb{Z}/25\mathbb{Z})^\times : \{76a_2/100\} \in [1/4, 3/4)\}$ has $|C| = 10$.

**Step 3: Squared magnitudes.** For $r \in (\mathbb{Z}/25\mathbb{Z})^\times$, define the ratio distribution $R_C(r) = \#\{(a,b) \in C \times C : ba^{-1} \equiv r \pmod{25}\}$, and analogously $R_{\mathrm{SG}}(r)$ for the signed diagonal set $\{1^+, 9^+, 3^-, 7^-\}$. Then $|S_G|^2 = 4\sum_r R_{\mathrm{SG}}(r)\,\psi(r)$ and $|B_{1,\overline{\chi}}|^2 = \sum_r R_C(r)\,\psi(r)$.

**Step 4: The even projection.** Let $f(r) = R_C(r) - 4\,R_{\mathrm{SG}}(r)$. Direct computation of all 20 coprime values of $r \in (\mathbb{Z}/25\mathbb{Z})^\times$ shows that the even part $f_+(r) = (f(r) + f(25-r))/2$ depends only on $r \bmod 5$: $f_+ = 2$ for $r \equiv \pm 1$ and $f_+ = 8$ for $r \equiv \pm 2$.

**Step 5: Kernel orthogonality.** The kernel of $(\mathbb{Z}/25\mathbb{Z})^\times \to (\mathbb{Z}/5\mathbb{Z})^\times$ is a subgroup of order 5. Any primitive character modulo 25 is nontrivial

on this kernel. Therefore any function depending only on $r$ mod 5 is orthogonal to all primitive characters.

**Step 6: Assembly.** Since $\psi$ is an even character ($\psi(-1) = 1$), only the even part of $f$ contributes: $\sum_r f(r)\,\psi(r) = \sum_r f_+(r)\,\psi(r) = 0$. Therefore $|B_{1,\overline{\chi}}|^2 = |S_G|^2$ for all 8 characters.

For base 6 ($m = 36 = 4 \times 9$): the CRT decomposition is over $(\mathbb{Z}/4\mathbb{Z})^\times \times (\mathbb{Z}/9\mathbb{Z})^\times$, giving 2 primitive odd characters. The even projection is constant on $(\mathbb{Z}/9\mathbb{Z})^\times$, and kernel orthogonality is immediate. □

**Corollary 5.5** (Fourth-moment identity)**.** *For bases $b \in \{6, 10\}$ and all primitive odd characters:*

$$|\widehat{c}(\chi)|^2 = \frac{b^4}{\pi^4\,\phi(m)^2}\,|L(1,\chi)|^4.$$

*The collision weights are a fourth power of the $L$-value.*

*Proof.* By Theorem 5.3: $|\widehat{c}|^2 = (b/\pi\varphi)^2\,|L|^2\,|S_G|^2 = (b/\pi\varphi)^2\,|L|^2\,|B_{1,\overline{\chi}}|^2 = (b^2/\pi^2\varphi(m))^2\,|L|^4$. □

**Observation 5.6.** The same fourth-moment identity holds at base 12, verified computationally for all 8 primitive odd characters modulo 144.

*Remark* 5.7. The identity holds for $b \in \{6, 10, 12\}$ and, verified computationally, no other base up to 60. For prime bases $b \le 5$, the ratio $|S_G|/|B_{1,\overline{\chi}}|$ is constant across characters but not equal to 1. For $b \ge 7$, the ratio varies. What distinguishes these three bases is open.

## 6. The Algebraic Structure of the Weights

The collision weights are not rational numbers.

**Theorem 6.1** (Weight decomposition in $\mathbb{Q}(\sqrt{5})$)**.** *For the 4 conjugate pairs of primitive odd characters modulo 100, the squared diagonal sums are:*

$$|S_G(\chi_k)|^2 = A_k + 2\epsilon_k\sqrt{5},$$

*where $(A_k, \epsilon_k) \in \{(14, +1),\ (10, +1),\ (14, -1),\ (10, -1)\}$ for $k = 2, 4, 6, 8$. The irrational parts cancel: $\sum_k 2\epsilon_k = 0$.*

*Proof.* The diagonal reduction gives $|S_G(\chi)|^2 = 4\,|\sum_{j\in\{1,3,7,9\}} \overline{\chi}(j)|^2$. For $\chi = \chi_4 \times \chi_{25}^k$, the character values involve 5th roots of unity, and the squared magnitude evaluates through $\cos(2\pi/5) = (\sqrt{5}-1)/4$. The cancellation follows from $\epsilon_k = -\epsilon_{20-k}$. □

The $\sqrt{5}$ arises from the odd prime factor of the base: $10 = 2 \times 5$. For base $b = 2p$ with $p \ge 5$ an odd prime, the collision weights live in $\mathbb{Q}(\sqrt{p})$: base 10 gives $\mathbb{Q}(\sqrt{5})$ (8 characters, 4 complement pairs),

base 14 gives $\mathbb{Q}(\sqrt{7})$ (18 characters). The structure requires at least two complement pairs ($\varphi(b^2)/2 \geq 4$) for the $\sqrt{p}$ component to distinguish them. For base 6 (2 characters, one pair), the weights are rational. The consequences for zero avoidance are developed in [4].

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